

Daniel Augustin

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Education

Northwestern University

2025 – 2026

Master of Science in Robotics

University of Massachusetts – Amherst

2019 – 2023

B.S. Triple Degree: Computer Science, Mechanical Engineering & Applied Mathematics Minor: Business

Professional Experience

Amazon — Redshift SDE

Jun 2023 – Aug 2024

California

- Designed and delivered new features for AWS Redshift, working across EC2 clusters, distributed databases, and internal DevOps pipelines.
- Built and integrated REST APIs with AWS Step Functions to automate customer data retrieval and processing workflows.
- Leveraged CloudWatch to monitor EC2, CPU, Amazon RDS, and EBS resources; configured alarms and log-based dashboards to improve system observability.
- Operated in a large-scale Agile environment, contributing to sprint planning, code review, and cross-team technical design sessions.

Amazon — AWS SDE Intern

Jun 2022 – Sep 2022

Virginia

- Worked across EMR, EC2, S3, Lambda, CloudFormation, EventBridge, CloudWatch, and developer tooling.
- Designed and implemented new data partitioning strategies for customer databases, improving query performance at scale.
- Delivered technical documentation and insights to support team decision-making.

Projects

Domino Hands

ROS2, Python, OpenCV, MoveIt2, RealSense, Franka Emika, Linux

- Built a fully autonomous robot system that detects, picks up, and precisely places dominoes into configurable patterns using a Franka Emika robot arm.
- Developed an RGB-D pose estimation pipeline for small-scale object detection using depth data and OpenCV bounding boxes.
- Implemented a 3-stage collision-aware manipulation pipeline (pickup → staging → placement) to navigate tight spatial constraints in dense domino arrays.
- Integrated joint effort monitoring for force-feedback placement on non-uniform surfaces; orchestrated a multi-node ROS2 system across MoveIt, RealSense, and custom CV nodes.

Tennis Ball Retrieval Robot

ROS2, Python, C++, YOLOv8, OpenCV, ZED 2i, RealSense, Jetson Orin Nano

- Designed and built an autonomous mecanum-wheel robot that detects, chases, and retrieves tennis balls on a court, returning them to a designated drop zone.
- Trained a custom YOLOv8 model on annotated court footage for robust ball, home landmark, and net strap detection across varied lighting conditions.
- Implemented a dual-camera perception stack (ZED 2i far-range, RealSense close-range) with seamless handoff across a Jetson Orin Nano + Raspberry Pi distributed compute setup.
- Engineered a 7-state autonomous mission state machine with ghost-frame-robust ball stillness detection and drop zone exclusion logic.

Skills

Programming	Python, C++, C, Java, JavaScript, TypeScript, MATLAB, R
Robotics / CV	ROS2, OpenCV, YOLOv8, MoveIt2, RealSense, ZED 2i, RVIZ, Gazebo, Taichi
Cloud / DevOps	AWS (EC2, S3, Lambda, EMR, Redshift), CloudWatch, CloudFormation, EventBridge, Step Functions, SQL
Tools	SOLIDWORKS, ANSYS, Git, JetBrains IDEs
Languages	English (native), Spanish (fluent), French (fluent), Haitian Creole (fluent)